Grouped predictor importance

Atmospheric Demand & Heat | Hydrological Supply | Soil Moisture Dynamics | Thermal & Radiation Regimes | Memory & Spatial Baseline

(a) ElasticNet

Vapor Pressure Deficit (VPD) 10.4%
Reference Evapotranspiration (ET0) 2.3%
Moderate Heat Days (>=30C)
Extreme Heat Days (>=35C) 0.3%
Extreme Degree Days (>30C) 12.6%
Cumulative Precipitation (P) 5.3%
Climatic Water Balance (P - ET0) 3.1%
Topsoil Moisture (0-7 cm) 11.3%
Mid Subsoil Moisture (7-28 cm) 5.0%
Deep Subsoil Moisture (28-100 cm) 3.5%
Root-Zone Soil Moisture 1.2%
Mean Temperature (T_mean)
Max Temperature (T_max) 5.5%
Min Temperature (T_min) 2.5%
Growing Degree Days (GDD) 0.4%
Dewpoint Temperature (T_dew) 27.9%
Surface Solar Radiation (SSRD) 6.7%
Lagged Yield Memory (ε_{t-1}) 0.7%
Spatial Coordinates (Lat / Lon) 1.4%

0% 5% 10% 15% 20% 25% 30%
Information Share (%)

(b) XGBoost

18.9%
8.7%
0.9%
0.3%
8.4%
8.9%
4.5%
5.9%
0.9%
3.6%
3.7%
3.6%
7.0%
3.0%
9.2%
8.0%
4.1%

0% 5% 10% 15% 20% 25% 30%
Information Share (%)

(c) LSTM (Surrogate)

3.7%
4.5%
0.4%
0.5%
4.1%
4.8%
24.9%
0.3%
0.4%
0.8%
1.7%
0.3%
1.2%
18.8%
8.2%
21.2%
3.9%

0% 5% 10% 15% 20% 25% 30%
Information Share (%)